

WDDA FS 2026: Guide for Problem Set 7

2026-05-20

1 Introduction

This guide provides detailed explanations for the exercises in WDDA FS 2026 Problem Set 7. The set focuses on **inference in regression models** – that is, confidence intervals, significance tests, and prediction intervals. We learn how reliable estimated coefficients and predictions are and how to test hypotheses about model parameters.

2 Exercise 1: Advertising – Confidence intervals and bootstrap

Task: Analyse the advertising dataset and compare classical and bootstrap confidence intervals for the coefficients of simple and multiple regressions. Examine the significance of the predictors and perform a model comparison.

2.1 Step 1: Read data and fit a simple model (a)

```
Advertising <- read_excel("../data/WDDA_07.xlsx", sheet = "Advertising")
head(Advertising)
```

```
## # A tibble: 6 x 4
##       TV radio newspaper sales
##   <dbl> <dbl>   <dbl> <dbl>
## 1 230.   37.8     69.2  22.1
## 2  44.5   39.3     45.1  10.4
## 3  17.2   45.9     69.3   9.3
## 4 152.   41.3     58.5  18.5
## 5 181.   10.8     58.4  12.9
## 6   8.7  48.9      75    7.2
```

```
mod_tv <- lm(sales ~ TV, data = Advertising)
summary(mod_tv)
```

```
##
## Call:
## lm(formula = sales ~ TV, data = Advertising)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.3860 -1.9545 -0.1913  2.0671  7.2124
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.032594   0.457843   15.36  <2e-16 ***
## TV           0.047537   0.002691   17.67  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 3.259 on 198 degrees of freedom
## Multiple R-squared: 0.6119, Adjusted R-squared: 0.6099
## F-statistic: 312.1 on 1 and 198 DF, p-value: < 2.2e-16
```

Interpretation:

- **Intercept** ($\beta_0 \approx 7.03$): expected sales when TV = 0.
- **TV coefficient** ($\beta_1 \approx 0.048$): for each additional 1\$ of TV spending, sales increase by about 0.048 units on average.
- Both coefficients are highly significant ($p < 0.001$).

2.2 Step 2: Classical confidence interval (b)

```
confint(mod_tv)
```

```
##                2.5 %      97.5 %
## (Intercept) 6.12971927 7.93546783
## TV          0.04223072 0.05284256
```

Interpretation: The 95% confidence interval for the TV coefficient is roughly between 0.042 and 0.053 – i.e. we are 95% confident that the true effect of TV spending on sales lies in this range.

2.3 Step 3: Bootstrap confidence interval (c)

```
set.seed(123)
simcoef <- do(5000) * coef(lm(sales ~ TV, data = resample(Advertising)))

CI_intercept <- quantile(simcoef$Intercept, c(0.025, 0.975))
CI_tv <- quantile(simcoef$TV, c(0.025, 0.975))

cat("Bootstrap 95% CI for Intercept:", round(CI_intercept, 3), "\n")
```

```
## Bootstrap 95% CI for Intercept: 6.391 7.697
```

```
cat("Bootstrap 95% CI for TV:      ", round(CI_tv, 3), "\n")
```

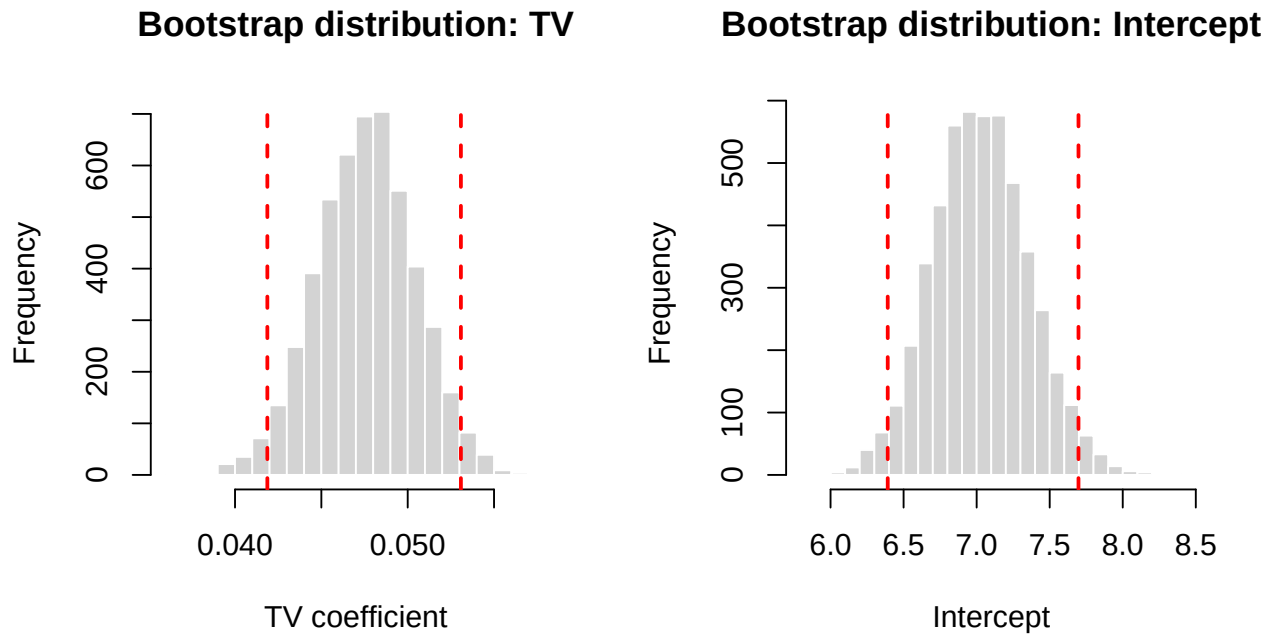
```
## Bootstrap 95% CI for TV:      0.042 0.053
```

Comparison: The bootstrap intervals agree closely with the classical intervals. This is expected when the classical assumptions (normality of residuals, constant variance) are reasonably well met.

2.4 Step 4: Visualising the bootstrap distribution (d)

```
par(mfrow = c(1, 2))
hist(simcoef$TV, main = "Bootstrap distribution: TV", xlab = "TV coefficient",
     col = "lightgray", border = "white", breaks = 25)
abline(v = quantile(simcoef$TV, c(0.025, 0.975)), col = "red", lty = 2, lwd = 2)

hist(simcoef$Intercept, main = "Bootstrap distribution: Intercept", xlab = "Intercept",
     col = "lightgray", border = "white", breaks = 25)
abline(v = quantile(simcoef$Intercept, c(0.025, 0.975)), col = "red", lty = 2, lwd = 2)
```



```
par(mfrow = c(1, 1))
```

Assessment: Both distributions are approximately bell-shaped (central limit theorem). The red lines mark the 95% confidence bounds.

2.5 Step 5: Multiple regression (e)

```
mod_full <- lm(sales ~ TV + radio + newspaper, data = Advertising)
summary(mod_full)
```

```
##
## Call:
## lm(formula = sales ~ TV + radio + newspaper, data = Advertising)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.8277 -0.8908  0.2418  1.1893  2.8292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.938889   0.311908   9.422  <2e-16 ***
## TV           0.045765   0.001395  32.809  <2e-16 ***
## radio        0.188530   0.008611  21.893  <2e-16 ***
## newspaper    -0.001037   0.005871  -0.177    0.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.686 on 196 degrees of freedom
## Multiple R-squared:  0.8972, Adjusted R-squared:  0.8956
## F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

Interpretation:

- TV and radio are highly significant ($p < 0.001$).
- newspaper is **not significant** ($p \approx 0.86$) – its effect is statistically indistinguishable from zero.

- The F-test ($p < 2.2 \cdot 10^{-16}$) shows that the model overall explains sales significantly.

2.6 Step 6: Model comparison via ANOVA F-test (f)

```
mod_red <- lm(sales ~ TV + radio, data = Advertising)
anova(mod_red, mod_full)
```

```
## Analysis of Variance Table
##
## Model 1: sales ~ TV + radio
## Model 2: sales ~ TV + radio + newspaper
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     197 556.91
## 2     196 556.83  1  0.088717 0.0312 0.8599
```

Interpretation:

- The F-test compares the reduced model (TV + radio) with the full model.
- $p > 0.05$ means: **newspaper** does not contribute significant additional explanatory power.
- **Conclusion:** we can drop **newspaper** from the model without losing meaningful explanatory power.

2.7 Step 7: Prediction with confidence and prediction intervals (g)

```
new_values <- data.frame(TV = 200, radio = 30)

ci_mean <- predict(mod_red, newdata = new_values, interval = "confidence")
pi_obs <- predict(mod_red, newdata = new_values, interval = "prediction")

cat("Confidence interval for mean sales:\n")
```

```
## Confidence interval for mean sales:
```

```
print(ci_mean)
```

```
##           fit          lwr          upr
## 1 17.71189 17.41904 18.00474
```

```
cat("\nPrediction interval for an individual observation:\n")
```

```
##
```

```
## Prediction interval for an individual observation:
```

```
print(pi_obs)
```

```
##           fit          lwr          upr
## 1 17.71189 14.38321 21.04057
```

Interpretation:

- The **confidence interval** (lwr, upr in ci_mean) refers to the **expected value** of sales at the given predictor values – i.e. the mean over many hypothetical observations.
- The **prediction interval** is wider because it additionally accounts for the spread of **individual** observations around the regression surface.

3 Exercise 2: Diamond Rings – Inference and prediction

Task: For the Diamond Rings dataset, investigate the relationship between weight and price. Compute confidence and prediction intervals and test the significance of the slope.

3.1 Step 1: Data and model (a)

```
diamonds <- read_excel("../data/WDDA_07.xlsx", sheet = "Diamonds Rings")
names(diamonds) <- c("weight", "price")
head(diamonds)
```

```
## # A tibble: 6 x 2
##   weight price
##   <dbl> <dbl>
## 1  0.17  355
## 2  0.16  328
## 3  0.17  350
## 4  0.18  325
## 5  0.25  642
## 6  0.16  342
```

```
mod_dr <- lm(price ~ weight, data = diamonds)
summary(mod_dr)
```

```
##
## Call:
## lm(formula = price ~ weight, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -85.159 -21.448  -0.869  18.972  79.370
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -259.63      17.32   -14.99  <2e-16 ***
## weight       3721.02      81.79    45.50  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.84 on 46 degrees of freedom
## Multiple R-squared:  0.9783, Adjusted R-squared:  0.9778
## F-statistic: 2070 on 1 and 46 DF, p-value: < 2.2e-16
```

Interpretation:

- **Slope (≈ 3721):** for each additional carat, the price increases on average by about 3721 SGD.
- The t-value for **weight** is large, p is extremely small – the effect is highly significant.

3.2 Step 2: Confidence interval for the slope (b)

```
confint(mod_dr)
```

```
##              2.5 %    97.5 %
## (Intercept) -294.487 -224.7649
## weight      3556.398 3885.6513
```

Interpretation: The 95% confidence interval for the slope is narrow and clearly above zero – the relationship is statistically well established.

3.3 Step 3: Bootstrap confidence interval (c)

```
set.seed(42)
simcoef_dr <- do(5000) * coef(lm(price ~ weight, data = resample(diamonds)))

CI_weight <- quantile(simcoef_dr$weight, c(0.025, 0.975))
cat("Bootstrap 95% CI for slope (weight):", round(CI_weight, 1), "\n")
```

```
## Bootstrap 95% CI for slope (weight): 3570.4 3870.6
```

Comparison: Bootstrap and classical intervals are very similar in this case.

3.4 Step 4: Prediction for a new ring (d)

```
newr <- data.frame(weight = 0.25)

ci_mean <- predict(mod_dr, newdata = newr, interval = "confidence")
pi_obs <- predict(mod_dr, newdata = newr, interval = "prediction")

cat("Mean expected price (95% CI):\n")
```

```
## Mean expected price (95% CI):
```

```
print(ci_mean)
```

```
##          fit          lwr          upr
## 1 670.6303 658.6925 682.5681
```

```
cat("\nPrediction interval for a single ring (95% PI):\n")
```

```
##
## Prediction interval for a single ring (95% PI):
```

```
print(pi_obs)
```

```
##          fit          lwr          upr
## 1 670.6303 605.4364 735.8242
```

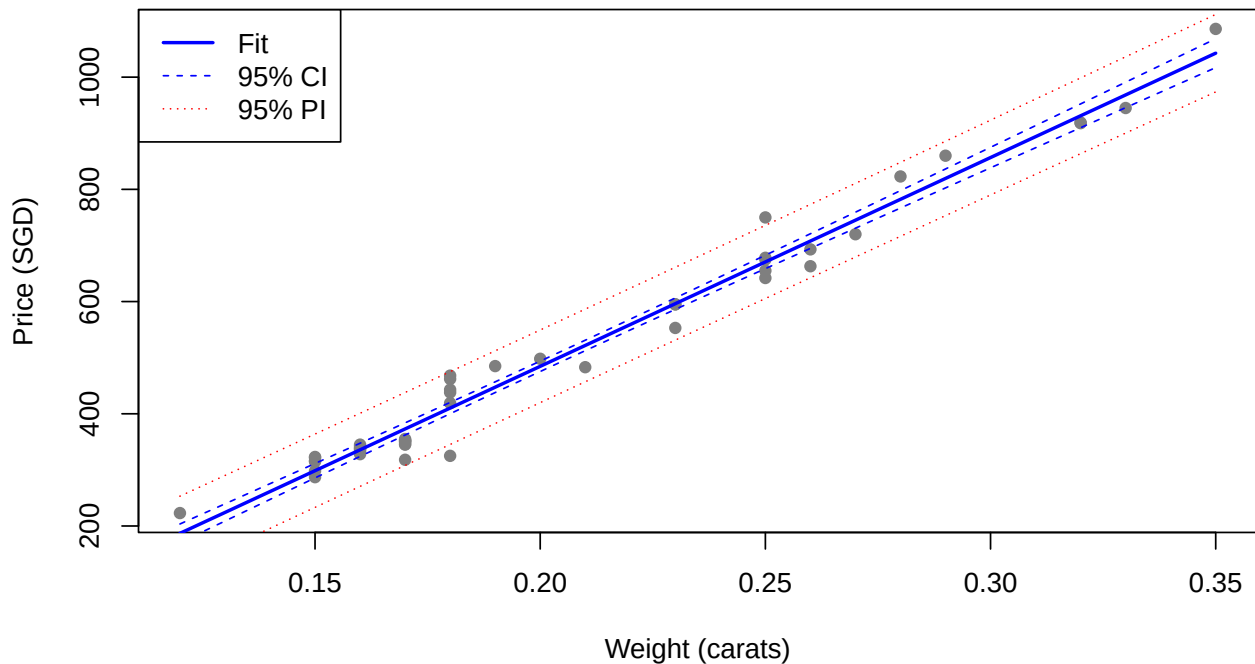
Interpretation: The prediction interval is wider than the confidence interval because it also accounts for the individual variability around the regression line.

3.5 Step 5: Visualisation with intervals (e)

```
grid <- data.frame(weight = seq(min(diamonds$weight), max(diamonds$weight), length.out = 100))
ci <- predict(mod_dr, newdata = grid, interval = "confidence")
pi <- predict(mod_dr, newdata = grid, interval = "prediction")

plot(diamonds$weight, diamonds$price, pch = 16, col = "gray50",
     xlab = "Weight (carats)", ylab = "Price (SGD)",
     main = "Regression with 95% CI and PI")
lines(grid$weight, ci[, "fit"], col = "blue", lwd = 2)
lines(grid$weight, ci[, "lwr"], col = "blue", lty = 2)
lines(grid$weight, ci[, "upr"], col = "blue", lty = 2)
lines(grid$weight, pi[, "lwr"], col = "red", lty = 3)
lines(grid$weight, pi[, "upr"], col = "red", lty = 3)
legend("topleft", legend = c("Fit", "95% CI", "95% PI"),
     col = c("blue", "blue", "red"), lty = c(1, 2, 3), lwd = c(2, 1, 1))
```

Regression with 95% CI and PI



Assessment: The PI band (red) is clearly wider than the CI band (blue). Both are narrowest at the centre of the data and widen towards the edges.

4 Exercise 3: SAT – Multiple regression and the F-test

Task: For the SAT dataset, examine which educational variables best explain the average SAT score of a US state. Perform model comparisons using an F-test.

4.1 Step 1: Read and inspect the data (a)

```
sat <- read_excel("../data/WDDA_07.xlsx", sheet = "SAT")
head(sat)
```

```
## # A tibble: 6 x 8
##   State      expend ratio salary  frac verbal  math  sat
##   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Alabama      4.40  17.2  31.1     8   491   538  1029
## 2 Alaska       8.96  17.6  48.0    47   445   489   934
## 3 Arizona      4.78  19.3  32.2    27   448   496   944
## 4 Arkansas     4.46  17.1  28.9     6   482   523  1005
## 5 California   4.99  24    41.1    45   417   485   902
## 6 Colorado     5.44  18.4  34.6    29   462   518   980
```

```
summary(sat[, c("expend", "ratio", "salary", "frac", "sat")])
```

```
##      expend      ratio      salary      frac
##  Min.   :3.656   Min.   :13.80   Min.   :25.99   Min.    : 4.00
##  1st Qu.:4.882   1st Qu.:15.22   1st Qu.:30.98   1st Qu.: 9.00
##  Median :5.768   Median :16.60   Median :33.29   Median :28.00
##  Mean   :5.905   Mean   :16.86   Mean   :34.83   Mean   :35.24
```

```
## 3rd Qu.:6.434 3rd Qu.:17.57 3rd Qu.:38.55 3rd Qu.:63.00
## Max. :9.774 Max. :24.30 Max. :50.05 Max. :81.00
## sat
## Min. : 844.0
## 1st Qu.: 897.2
## Median : 945.5
## Mean : 965.9
## 3rd Qu.:1032.0
## Max. :1107.0
```

Description:

- **expend**: per-pupil education spending (in 1000\$)
- **ratio**: pupil-teacher ratio
- **salary**: average teacher salary (in 1000\$)
- **frac**: fraction of students taking the SAT (in %)
- **sat**: average SAT score

4.2 Step 2: Simple model (b)

```
mod1 <- lm(sat ~ expend, data = sat)
summary(mod1)
```

```
##
## Call:
## lm(formula = sat ~ expend, data = sat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -145.074  -46.821    4.087   40.034  128.489
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1089.294     44.390   24.539 < 2e-16 ***
## expend      -20.892      7.328   -2.851  0.00641 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 69.91 on 48 degrees of freedom
## Multiple R-squared:  0.1448, Adjusted R-squared:  0.127
## F-statistic: 8.128 on 1 and 48 DF, p-value: 0.006408
```

Surprise: The slope on **expend** is **negative**! Higher education spending is associated with lower SAT scores? This looks paradoxical – a hint at a confounding variable.

4.3 Step 3: Multiple regression (c)

```
mod2 <- lm(sat ~ expend + ratio + salary + frac, data = sat)
summary(mod2)
```

```
##
## Call:
## lm(formula = sat ~ expend + ratio + salary + frac, data = sat)
##
## Residuals:
```



```
##      Min      1Q  Median      3Q      Max
## -90.531 -20.855  -1.746  15.979  66.571
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1045.9715    52.8698  19.784 < 2e-16 ***
## expend      4.4626     10.5465   0.423  0.674
## ratio      -3.6242     3.2154  -1.127  0.266
## salary      1.6379     2.3872   0.686  0.496
## frac       -2.9045     0.2313 -12.559 2.61e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.7 on 45 degrees of freedom
## Multiple R-squared:  0.8246, Adjusted R-squared:  0.809
## F-statistic: 52.88 on 4 and 45 DF,  p-value: < 2.2e-16
```

Interpretation:

- After controlling for `frac` (fraction of students participating), **the sign of many coefficients flips**.
- `frac` is highly significant and negative: states where more students take the SAT have lower average scores (self-selection).
- `expend` and `salary` now have positive signs – the true effect of additional resources is actually positive once we control for the selection effect.

4.4 Step 4: F-test of the overall model (d)

```
f <- summary(mod2)$fstatistic
cat("F statistic:", round(f["value"], 2),
    " (df1 =", f["numdf"], ", df2 =", f["dendf"], ")\\n")
```

```
## F statistic: 52.88 (df1 = 4 , df2 = 45 )
```

Interpretation: The F-test checks the global null hypothesis that **all** slopes are zero. A high F-value with a small p rejects this hypothesis – at least one predictor explains significant variance.

4.5 Step 5: ANOVA comparison of nested models (e)

```
mod_small <- lm(sat ~ frac, data = sat)
mod_large <- lm(sat ~ expend + ratio + salary + frac, data = sat)
anova(mod_small, mod_large)
```

```
## Analysis of Variance Table
##
## Model 1: sat ~ frac
## Model 2: sat ~ expend + ratio + salary + frac
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1      48 58433
## 2      45 48124  3    10309 3.2133 0.03165 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- Compares the reduced model (only `frac`) with the full model.

- A significant p -value would mean that the extra predictors jointly contribute significantly to the explanation.
- If $p > 0.05$: the extra variables provide no significant gain – **frac** alone captures most of the variation.

4.6 Step 6: Prediction for a hypothetical state (f)

```
hypothetical <- data.frame(expend = 6, ratio = 16, salary = 35, frac = 30)
pred <- predict(mod_large, newdata = hypothetical, interval = "prediction")
print(pred)
```

```
##          fit          lwr          upr
## 1 984.9521 918.1638 1051.74
```

Interpretation: Expected SAT score plus 95% prediction interval for a state with the specified characteristics.

5 Summary

This problem set has shown:

1. **Classical and bootstrap confidence intervals** yield very similar results when model assumptions hold. Bootstrap is more robust when assumptions are violated.
2. **t-tests** check individual coefficients, **F-tests** check the overall model or model comparisons.
3. **Confidence vs. prediction intervals**: the CI refers to the expected value, the PI to a single new observation – the PI is always wider.
4. **Multiple regression** allows estimating effects of individual predictors controlling for other variables – important for avoiding confounding (SAT example).
5. **ANOVA F-tests** compare nested models and help with model selection: simpler models are preferred unless they are significantly worse.